

- 12** A satellite of mass 50 kg moves from a point where the gravitational potential due to the Earth is -30 MJ kg^{-1} to another point where the gravitational potential is -70 MJ kg^{-1} .

During this change of position, it has moved

- A** closer to the Earth and lost 2000 MJ of potential energy.
- B** closer to the Earth and lost 40 MJ of potential energy.
- C** further from the Earth and gained 2000 MJ of potential energy.
- D** further from the Earth and gained 40 MJ of potential energy.

- 13** A fixed mass of ideal gas undergoes changes in pressure p and volume V starting with